

Practical tasks

— ruler-and-compass constructions

The national “practical tasks” lesson, carrying the Cambridge Stage 9 constructions the national programme does not cover.

LESSON 16

1 × 40 MIN

GEOMETRY 8, TEMA 12

STAGE 9 · 5.4 [CAMBRIDGE INSERT]

LESSON CLOCK

4'

10'

10'

14'

2'

Warm-up	4 min
Explanation	10 min
Construct	10 min
Practice	14 min
Homework	2 min

LEARNING OBJECTIVES

- Construct the perpendicular bisector of a segment and explain why it works.
- Construct the bisector of an angle.
- Construct a perpendicular from a point to a line, and an angle of 60° and 30°.
- Use a construction to solve a practical placement problem.

TEXTBOOK

National	Tema 12, pp. 29–32
Cambridge	Learner’s Book pp. 119–123
Workbook	Workbook 5.4

01

Explanation

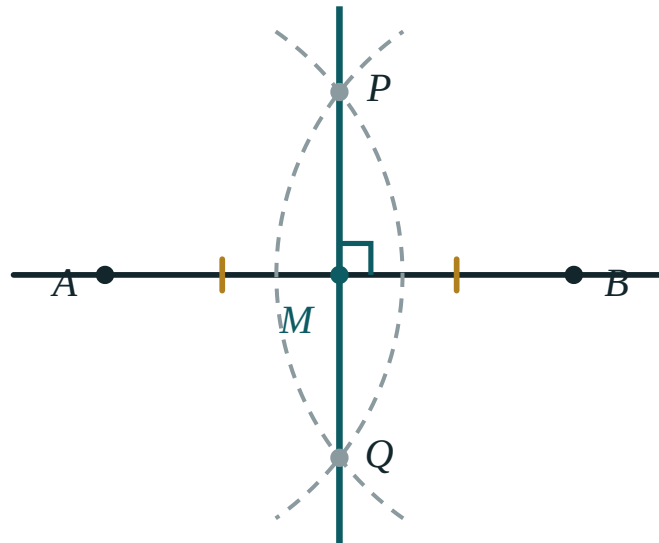
The rules of the game

ALLOWED

A straight edge, to draw a line through two points, and compasses, to draw a circle of a chosen radius about a chosen point. **Not allowed:** measuring lengths with the ruler’s scale, or angles with a protractor.

Leave every construction arc on the page. The arcs are the working, and in Cambridge papers they carry the marks — a correct figure with the arcs rubbed out scores less than a slightly wobbly one with them showing.

Construction 1 — the perpendicular bisector of AB

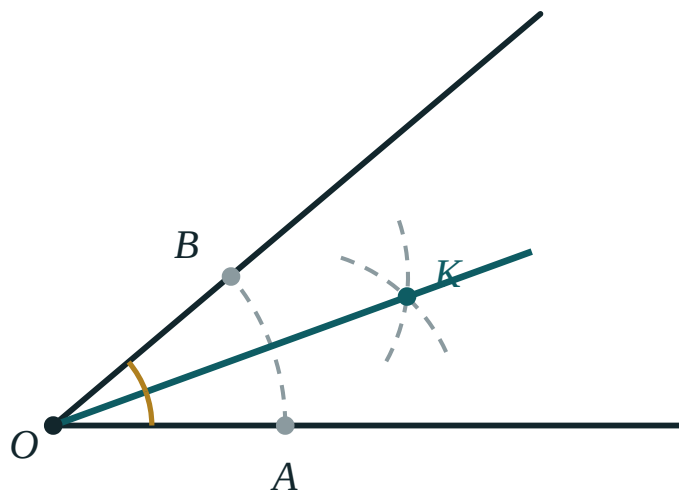


Equal radii from A and from B meet at P and Q. The line PQ is the perpendicular bisector of AB.

1. Open the compasses to more than half of AB .
2. Draw an arc centred at A , and another of the **same radius** centred at B .
3. They meet at two points, P and Q . Join them.

Why it works. P and Q are each the same distance from A and from B , so both lie on the set of points equidistant from A and B — which is exactly the perpendicular bisector. The line also gives the **midpoint** M for free.

Construction 2 — the bisector of an angle



One arc from O cuts the arms at A and B ; equal arcs from A and B meet at K ; OK is the bisector.

1. With centre O draw an arc cutting the arms at A and B .

2. With the same radius, draw arcs centred at A and at B ; call their intersection K .

3. Draw OK .

Why it works. $OA = OB$ and $AK = BK$ and OK is common, so $\triangle OAK \cong \triangle OBK$ by SSS, and therefore $\angle AOK = \angle KOB$. ■

Two more you should be able to do

A perpendicular from a point P to a line ℓ . Draw an arc centred at P cutting ℓ at two points, then construct the perpendicular bisector of the segment between them — it passes through P .

An angle of 60° . Draw an arc centred at O cutting the line at A ; with the same radius, an arc centred at A cuts the first at B . Then $\triangle OAB$ is equilateral, so $\angle AOB = 60^\circ$. Bisect it for 30° , bisect again for 15° .

WHERE THESE ARE USED

Every one of these is a **locus**: the perpendicular bisector is the set of points equidistant from two points; the angle bisector is the set of points equidistant from two lines. That is why they answer questions like “where should the well be dug so that it is the same distance from both villages?”

02 Worked examples

EXAMPLE 1 Find the point equidistant from three villages A, B and C.

- 1 Construct the perpendicular bisector of AB .
Every point on it is equidistant from A and B.
- 2 Construct the perpendicular bisector of BC .
Every point on it is equidistant from B and C.
- 3 They meet at one point O .
 $OA = OB$ and $OB = OC$, so all three are equal.
- 4 O is the centre of the circle through A, B, C.
The circumcentre of the triangle.

ANSWER

The intersection of the perpendicular bisectors — the circumcentre.

EXAMPLE 2 Construct an angle of 45° using compasses only.

- 1 Construct a perpendicular to a line at a point O .
That gives 90° .
- 2 Bisect the right angle.
Two equal halves of 90° .
- 3 Each half is 45° .
No protractor was used at any stage.

ANSWER

Perpendicular, then bisect.

EXAMPLE 3 A path must be the same distance from two straight roads that cross. Where does it run?

- ① The set of points equidistant from two lines is the bisector of the angle between them.
- ② Two crossing roads make two pairs of vertically opposite angles.
So there are two bisectors.
- ③ The two bisectors are perpendicular to each other.
Both answers are valid paths.

ANSWER

Along either bisector of the angles between the roads.

03 **Interactive model**

Do these on the board with real compasses first, then use the step-through to check the reasoning.

Construction steps

Given a segment AB .

① Open the compasses to more than half of

AB .

Any
radius
under
half
gives
no
intersection.

② Arc centred at

A ; arc of the

same radius centred at

B .

The
equal
radius
is
the
whole
point.

③ They meet at

P and

Q . Draw the line

PQ .

4 $PA = PB$

and $QA = QB$

, so PQ

is A

the

locus

of

points

equidistant

from

and B

· It is perpendicular to AB and passes through its midpoint.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Compasses	Sirkul	Циркуль
Straight edge	Chizg‘ich	Линейка
Arc	Yoy	Дуга
Radius	Radius	Радиус
Perpendicular bisector	O‘rta perpendikulyar	Серединный перпендикуляр
Angle bisector	Burchak bissektisasi	Биссектриса угла
Locus	Geometrik o‘rin	Геометрическое место точек
Equidistant	Teng uzoqlikda	Равноудалённый
Circumcentre	Tashqi aylana markazi	Центр описанной окружности
Incentre	Ichki aylana markazi	Центр вписанной окружности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For the perpendicular bisector construction the compass radius must be:

exactly half of AB

more than half of AB

less than half of AB

any radius at all

The set of points equidistant from two points is:

a circle

the perpendicular bisector of the segment joining them

an angle bisector

a parabola

The set of points equidistant from two crossing lines is:

a circle

the perpendicular bisector

the pair of angle bisectors

a single point

To construct 30° you first construct:

45°

60° and bisect it

90° and trisect it

15° and double it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Construct the perpendicular bisector of a 6 cm segment.*

ANSWER Equal arcs from both ends; join the two intersections.

2. *Construct the midpoint of a 7 cm segment.*

ANSWER It is where the perpendicular bisector crosses.

3. *Construct the bisector of a 70° angle. What is each half?*

ANSWER 35°

4. *Construct an angle of 60° .*

ANSWER Equilateral triangle by equal compass widths.

5. *Construct an angle of 30° .*

ANSWER Construct 60° and bisect it.

6. *What is the locus of points equidistant from A and B?*

ANSWER The perpendicular bisector of AB.

7. *What is the locus of points 3 cm from a point O?*

ANSWER A circle of radius 3 cm centred at O.

Medium · the standard question

7 PROBLEMS

1. *Construct a perpendicular from a point P to a line ℓ .*

ANSWER Arc from P cutting ℓ twice, then the perpendicular bisector of that segment.

2. *Construct an angle of 45° .*

ANSWER 90° then bisect.

3. *Construct an angle of 15° .*

ANSWER $60^\circ \rightarrow 30^\circ \rightarrow 15^\circ$ by bisecting twice.

4. *Construct the circumcentre of a triangle.*

ANSWER Intersection of two perpendicular bisectors.

5. *Construct the incentre of a triangle.*

ANSWER Intersection of two angle bisectors.

6. *Explain why the angle-bisector construction works.*

ANSWER $\triangle OAK \cong \triangle OBK$ by SSS.

7. *Construct a triangle with sides 5 cm, 6 cm and 7 cm.*

ANSWER Draw the 7 cm base, then arcs of 5 cm and 6 cm from its ends.

Hard · stretch and reason

7 PROBLEMS

1. *A well must be equidistant from three houses. Where is it, and when is there no such point?*

ANSWER At the circumcentre; there is none if the three houses lie in a straight line.

2. *Construct an angle of 75° .*

ANSWER $60^\circ + 15^\circ$ — build 60° , then bisect 30° onto it.

3. *A treasure is 4 cm from point A and equidistant from B and C. Describe the construction.*

ANSWER Circle of radius 4 cm about A, intersected with the perpendicular bisector of BC — up to two positions.

4. *Prove that the perpendicular bisectors of the three sides of a triangle are concurrent.*

ANSWER The intersection of two of them is equidistant from all three vertices, so it lies on the third as well.

5. *Construct the tangent-length point: divide a 9 cm segment into 3 equal parts by construction.*

ANSWER Thales' construction: three equal steps on a ray, join the last to B, draw parallels.

6. *Construct an angle of 105° and explain your steps.*

ANSWER $90^\circ + 15^\circ$: build a perpendicular, then bisect 60° twice to get 15° and add it.

7. *A path must be equidistant from two roads and 5 cm from a crossroads. Find the possible positions.*

ANSWER Intersect the two angle bisectors with the circle of radius 5 cm — up to four points.

07 Homework

Homework — 5 constructions

Geometry 8, Tema 12, pp. 29–32, and Cambridge Learner's Book 5.4. Leave all arcs visible.

1. *Construct the perpendicular bisector of an 8 cm segment.*
2. *Construct the bisector of an angle of about 100° and measure each half to check.*
3. *Construct angles of 60° , 30° and 15° from the same line.*
4. *Construct a triangle with sides 4 cm, 6 cm and 7 cm, then its circumcentre.*

5. A house is to be built equidistant from two straight roads and 6 cm from a marked tree. Construct all possible positions.